

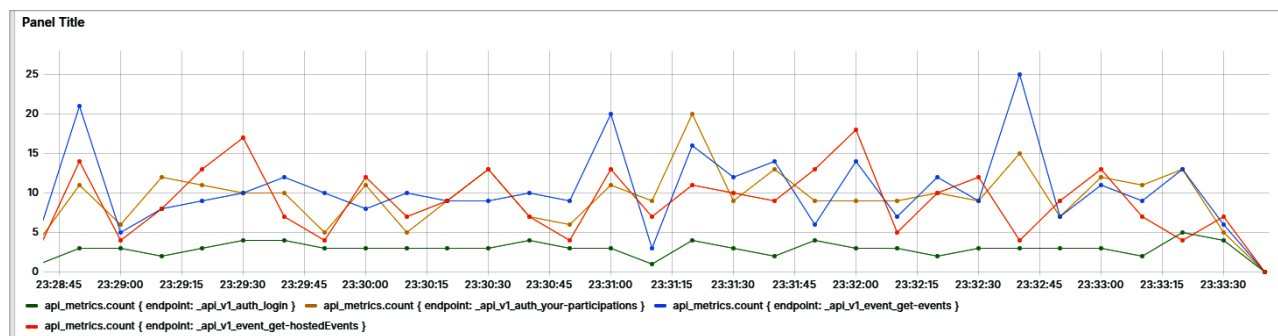


Figure 3: API count

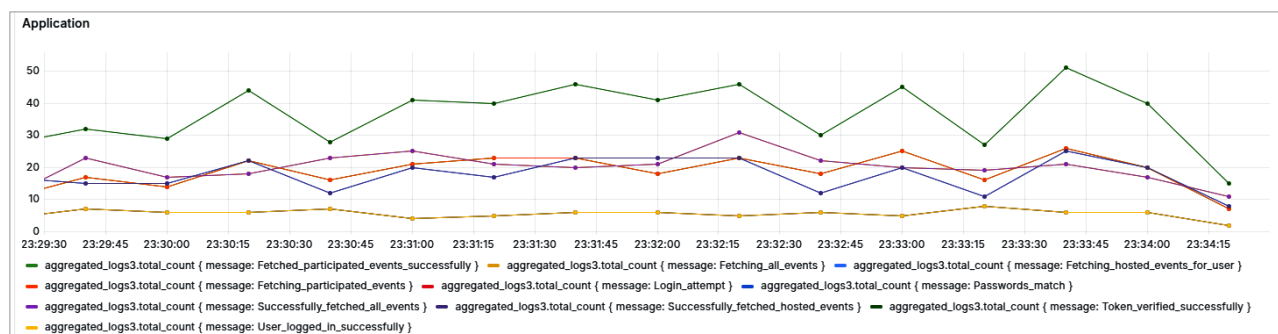


Figure 4: Log data

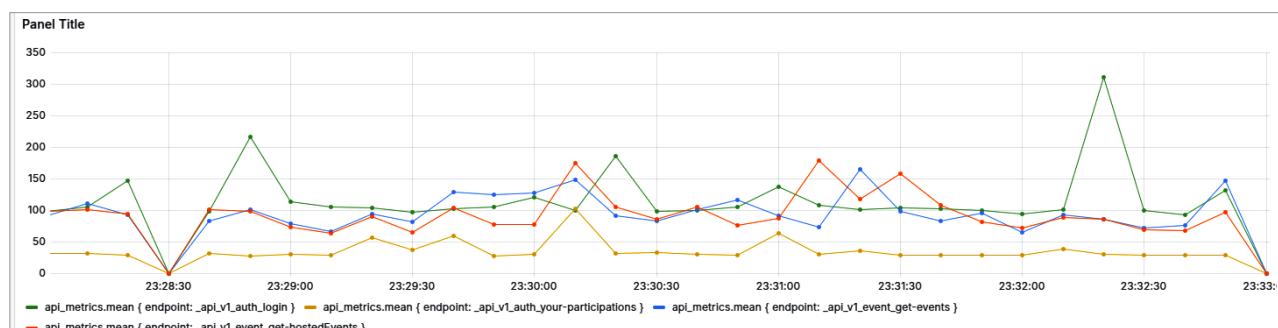


Figure 5: Mean response time

of system memory, including committed memory, available memory, used memory, free memory, cache, active memory, inactive memory, buffers, dirty memory, and percentage-based metrics like commit percentage and memory used percentage. The dominant part of the chart is occupied by committed memory, indicating a significant allocation of physical memory resources. This visualisation helps in understanding the memory consumption patterns of the system and identifying which components are using the most resources.

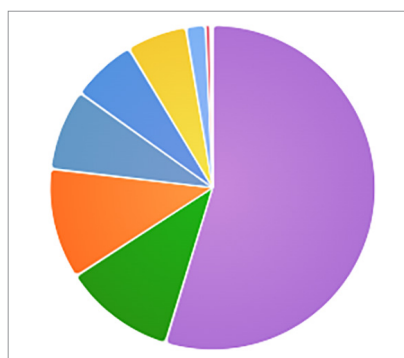


Figure 6: Memory metrics

Violet: Committed memory (KB)
Dark Green: Inactive memory (KB)
Orange: Available memory (KB)

Steel Blue: Used memory (KB)
Sky Blue: Cache (KB)
Yellow-Brown: Free memory (KB)
Teal: Active memory (KB)
Red: Buffers (KB)
Purple: Dirty memory (KB)
Green: Commit (%)
Gold: Memory used (%)

Challenges and solutions

Scalability: For scalability, the system has been created with a modular architecture, isolating concerns such as event management, authentication, and logging. The application of MongoDB enables easy scaling, particularly for